Advanced High School Math and the STEM Pipeline: Does Israel's "Give-Five" Reform Increase STEM Entry and Reduce Socioeconomic Gaps?

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1 Introduction

Israel's education system plays a central role in shaping the country's long run innovation capacity, yet participation in rigorous mathematics during high school has historically been highly uneven across communities. For more than a decade prior to 2015, the share of students completing the advanced 5 unit mathematics track declined steadily, particularly in weaker municipalities. This decline raised concerns within the Ministry of Education that large segments of the population were being shut out of selective university programs in engineering, computer science, and the physical sciences. Since these programs serve as the main entry point into Israel's high productivity occupations, limited access to advanced mathematics implied restricted access to some of the key drivers of economic growth and social mobility.

The Give Five program was launched to address this challenge. The initiative aimed to expand the availability of advanced mathematics by enabling small or under resourced schools to open new tracks, strengthening instruction through dedicated funding, and promoting participation through public campaigns and enrichment activities. Policymakers hoped that raising the number of high school graduates with strong quantitative preparation would eventually increase the diversity and size of the pool entering STEM fields in higher education. While the immediate goal was to increase the supply of 5 unit mathematics students, the broader ambition was to influence long term academic and career outcomes. Understanding whether this occurred is important because it speaks directly to the effectiveness of large

scale curricular interventions and the capacity of education policy to alter the composition of Israel's future STEM workforce.

This project examines whether exposure to Give Five during high school was followed by higher rates of STEM enrollment at the undergraduate level. The analysis focuses on municipalities rather than individuals, which allows the use of detailed administrative data that record both high school exam patterns and higher education enrollment by locality of residence. I construct a municipality by year dataset that links these two stages of the education system. On the high school side, I use information from national matriculation records to measure the intensity of the expansion in advanced mathematics within each locality. On the university side, I use annual publications from the Central Bureau of Statistics that report how many new first degree students from each locality enter STEM fields. This combination allows me to trace how changes in high school preparation within a community relate to subsequent patterns of STEM participation among its graduates.

With these data, the empirical strategy proceeds in two steps. First, it documents broad descriptive patterns that show large differences across socio economic groups in both exposure to the reform and participation in STEM. These differences highlight the need for a careful design, because high and low SES municipalities do not evolve in parallel over time. Second, the analysis estimates whether localities that experienced larger expansions in advanced mathematics also exhibited larger increases in STEM enrollment after the relevant cohorts reached university age. The main approach uses differences over time within localities, controlling for persistent characteristics and national trends. Additional specifications introduce continuous measures of exposure and test for sensitive issues such as differential pre trends.

The goal of the paper is to provide credible, locality level evidence on whether a national effort to expand high school mathematics translated into meaningful changes in the STEM pipeline. Beyond the specific case of Israel, the findings speak to a broader policy question: can broad based curricular reforms in secondary education influence the composition of students entering high stakes, high return fields in higher education.

2 Literature Review

2.1 Background: Math Education and Social Mobility in Israel

Israel's rapid economic growth has been driven by advances in human capital and the expansion of knowledge-intensive industries. The high-tech sector alone accounts for about one-fifth of Israel's GDP and over half of its exports, making the availability of highly skilled workers a central concern for sustained growth. Several reports have identified a shortage of qualified workers in science and technology fields, citing both limited participation in advanced STEM studies and uneven quality of instruction (Kandel, 2012; Zuk, 2014; Shaviv, 2018).

STEM education, and particularly advanced mathematics, serves as the main gateway to these high-productivity sectors. In Israel's education system, completion of the 5-unit mathematics matriculation exam is a prerequisite for admission to most selective STEM programs in higher education, which in turn are associated with higher wages and stronger labor-market outcomes. Yet participation in 5-unit mathematics declined sharply during the late 2000s and early 2010s. The Knesset Research and Information Center reported a drop of roughly 30 percent between 2006 and 2012. According to the Bank of Israel, between 2013 and 2016 the share of twelfth-grade students completing 5-unit mathematics fell from 16.1 percent to 10.9 percent, while participation in 4-unit mathematics remained steady and 3-unit participation increased. The decline affected all population groups but was especially pronounced among students from weaker socioeconomic backgrounds.

These patterns have broad implications for social mobility. OECD data show that educational attainment in Israel remains strongly linked to parental background: among adults aged 25-34, 63 percent of those with tertiary-educated parents also hold a tertiary degree, compared with only 25 percent among those whose parents did not complete secondary school. Limited educational mobility not only perpetuates inequality but also constrains the development of a skilled workforce essential for the Israeli economy. Expanding access to advanced mathematics in high school is therefore critical both for promoting inclusive economic growth and for improving equality of opportunity.

2.2 The "Give-Five" Reform and Previous Reforms

In August 2015 the Give Five national programme was launched by the Ministry of Education to reverse the decline in students taking the 5-unit matriculation exam in mathematics. The stated goal was to double the number of 5-unit mathematics matriculation students within four years, rising from approximately 9.7 thousand in 2014 to over 18 thousand, and to increase the share of 5-unit participants among all grade-12 students from roughly 10.8 percent toward 18 percent within ten years. [1]

The design of Give-Five included three key features. First, it targeted schools that previously struggled to offer the advanced mathematics track by relaxing the minimum class-size rule: from about 15 students to only six, thereby enabling small or disadvantaged schools to open a 5-unit mathematics class. Second, the programme provided additional resources: participating schools received budgets for roughly 15 weekly teaching hours per 12th-grade cohort, teacher training and certification programmes, and promotional activities including national mathematics marathons and teacher-student workshops. Third, a school selection and expansion approach was used: the 2015 launch included 108 high schools; by 2018 the number reached 406 high schools, representing about 39 percent of all ordinary secular high schools in Israel (excluding Haredi and special-education systems).

In parallel, the program built on and complemented earlier initiatives such as the Atuda ("Academic Reserve") science-technology program launched in 2011, which aimed to increase 5-unit mathematics and science participation in select schools and linking high school study with military and academic STEM tracks. While Atuda targeted more selective students and schools, Give-Five was broader with a focus on under-served schools and improving equity of access to advanced mathematics. Early evaluation from the Bank of Israel indicates that Give-Five increased the share of students taking the 5-unit exam by almost 3 percentage points (from about 14 percent) by 2017, without notable decline in pass rates or average scores. [2]

Overall, the reform aimed to expand access, raise status and quality, and enable smaller and weaker schools to offer advanced mathematics, thereby supporting both growth in high-skill STEM supply and greater educational equity across municipalities.

2.3 The STEM Pipeline: From High School to Higher Education

Note: Advanced math is a known "gatekeeper" for selective STEM programs in Israel), both via explicit prerequisites and through admissions bonuses.

An Israel Democracy Institute (IDI) analysis reports that 56.6 percent of graduates who passed 5-unit math pursued "realistic/scientific" tracks - computer science, mathematics/statistics, engineering, natural sciences, whereas the shares for 4- and 3-unit graduates were approximately 30 percent and 13 percent, respectively- evidence of a strong pipeline effect from advanced math to STEM majors [5]. The same analysis documents roughly a 50 percent rise in undergraduate math and computer science enrollment between 2012 and 2018 and notes capacity expansions funded by the Council for Higher Education (CHE). CHE's statistical portal provides multi-year tables on freshmen and total enrollment by field and by type of institution; these are crucial for separating research universities (selective STEM) from colleges (less selective on average) [3]. There is strong descriptive evidence that 5-unit math strongly predicts STEM enrollment and that STEM seats expanded overall in the late 2010s, but selectivity (university vs. college) and composition by origin/SES remain key unknowns.

2.4 What's Missing and Why the Proposed Study Fills It

The central gap in the existing literature is the missing link: we know that Israel's 2015 "Give Five" reform dramatically increased the number of students taking 5-unit math in high school, and we know that 5-unit math graduates are more likely to pursue STEM degrees. But there is no study that directly tracks whether this reform translated into greater access for disadvantaged students to selective STEM programs at the university level. Evaluations so far (Bank of Israel, Taub Center, IDI) have either measured only high-school participation, or reported overall STEM enrollment growth without breaking it down by socioeconomic background. As a result, it remains unclear whether the reform actually promoted social mobility or simply reinforced existing inequalities by disproportionately benefiting already strong students and schools. The Bank of Israel study's causal evidence ends at 5-unit take-up and subject participation; it does not track university admissions/enrollment by SES or selectivity [2]. Moreover, the Bank of Israel research (2019) acknowledges a data limitation: the first high-school cohorts affected by Give Five (2017 graduates) didn't yet begun their higher education, so the study did not examine the reform's impact on higher education. In addition, the analysis mainly categorizes students by Jewish/Arab sector; my goal is to distinguish low-SES origins within both Jewish and Arab populations, aiming to study social mobility effects regardless of ethnicity.

To sum up, the literature shows the reform worked at the high-school margin and that advanced math is a gateway to STEM. It does not yet show whether the 2015 policy improved equitable access to selective STEM programs for low-SES origins.

3 Data Description

This study combines detailed administrative and official statistical data to evaluate the effectiveness of Israel's 2015 "Give Five" reform in advanced mathematics at the locality level and to link high-school exposure to subsequent participation in selective STEM fields in higher education. The analysis draws on four main datasets, documenting the pipeline from secondary education to university entry.

The core of the analysis is based on Ministry of Education administrative data obtained through Freedom of Information (FOI) releases and the public "Bagrut map." These files provide school-level records of matriculation exams by subject, number of units, and year between 2013 and 2021. Each observation represents a unique school–subject–unit–year cell and includes information on the average test score, number of examinees, educational sector, and school characteristics such as district and type. After cleaning, the dataset includes 121,475 observations. Hebrew locality names were standardized and matched to CBS locality codes and socio-economic indices. Subjects were translated into English and normalized for spacing and encoding issues. Rows that could not be reliably assigned to a locality were dropped, primarily representing very small or special schools. The resulting

file, `exam_all_clean_final`, provides a complete view of the structure of high-school exam participation in Israel over nearly a decade.

Exploration of this dataset reveals wide variation in advanced mathematics participation across localities, with a clear upward shift following 2015. In some small municipalities, only a few students took the five-unit exam, leading to substantial year-to-year volatility in shares. These patterns motivate aggregation to the locality–year level to obtain stable measures of exposure.

From this foundation, I construct a second dataset, `math_5u_by_city_year`, which aggregates the Bagrut data to the locality–year level. For each municipality and year, I calculate the total number of math examinees and the number of students taking the five-unit version, then derive `share_5u`, the share of advanced-math examinees. The resulting panel includes 2,295 locality–year observations and forms the core "supply-side" measure of reform exposure.

To quantify exposure intensity, I take the average `share_5u` in the pre-reform period (2013–2014) and in the post-reform period (2016–2019) and compute the difference between them, Δ`share_5u`. This difference reflects how strongly each locality expanded its advanced-math participation following the reform. I then merge in socio-economic attributes from the CBS, including population size, socio-economic index, rank, and cluster (2013 and 2015 editions). Based on the distribution of Δ`share_5u`, localities are classified into high/medium/low exposure groups. The merged file, `exposure_by_town_with_SES`, provides both treatment intensity and socio-economic context for each locality, allowing analyses of heterogeneity and equity.

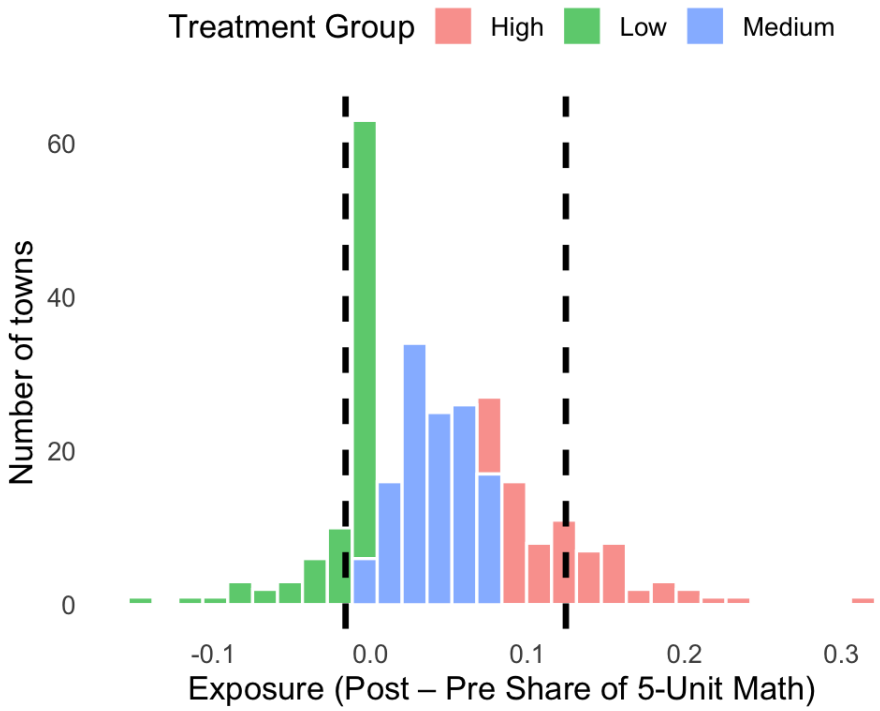


Figure 1: Distribution of Math 5-Unit Exposure Across Towns (2013–2019)

To measure outcomes, I use the CBS annual Students in Higher Education publications, which report the number of new first-degree students by field of study and by locality of residence at age 18 (interpreted as the student's home locality). This data spans academic years 2017-2024 and is organized at the locality-year-field level. The resulting dataset, NewStudentsV2, includes all major fields-engineering and architecture; mathematics, statistics, and computer science; physical and biological sciences; medicine and health; social sciences; humanities; law; and others. Because CBS revised field definitions in 2021, I aggregate them into stable STEM and non-STEM groups to ensure comparability over time. The main outcome variable is the share of new first-degree students entering STEM fields from each locality and year, defined as the number of new STEM students divided by the total number of new first-degree students. This measure provides a direct link between local high-school preparation and the academic paths chosen in higher education. Together, these datasets enable a comprehensive view of how the expansion of advanced-math study at the secondary level translated into changes in STEM participation at the tertiary level. The unit of analysis for the main regressions is locality-year, allowing for alignment between the exposure and outcome data through the CBS locality codes.

4 Data limitations

As with any administrative data, the construction process introduces several limitations. The most fundamental is that we do not have an official list of schools that participated directly in the "Give Five" program or the exact year each joined. Consequently, treatment intensity is inferred rather than assigned, using the observed increase in five-unit participation within each locality. This means the design measures realized exposure to reform incentives rather than formal program participation.

A second limitation is the absence of student-level identifiers. Because the data are aggregated, we cannot track individual students from high school to university. The interpretation is therefore intent-to-treat at the locality level: it estimates how an increase in the local prevalence of advanced-math study correlates with subsequent changes in STEM entry among graduates from that locality. Third, the process of mapping schools to localities required careful manual normalization. Certain regional councils and merged or hyphenated towns had inconsistent Hebrew naming conventions, and unmatched rows were excluded. These losses mainly affect small or rural localities. Additionally, in smaller towns, the number of examinees is small enough that even minor fluctuations can significantly alter the measured share of five-unit students, creating noisy estimates. Aggregation over multiple years and weighting by cohort size help mitigate this issue.

Finally, CBS reclassified several academic fields in 2021, which could complicate long-term comparisons. To maintain consistency, I define STEM as a fixed aggregate of four fields-engineering and architecture; mathematics, statistics, and computer science; physical sciences; and biological sciences-and include year fixed effects in all regressions to absorb any national-level shocks or definitional shifts.

In summary, the combined data provide an unusually detailed view of the educational pipeline from high school to university, spanning a decade of reform and reform-related outcomes. Despite the limitations-especially the absence of individual identifiers and precise program participation records-the locality-level framework remains well suited to identify whether areas that experienced greater expansion in advanced-math participation also saw stronger growth in STEM enrollment, and whether this relationship differs across socio-economic contexts.

5 Research Design

The empirical strategy uses a panel of localities observed annually during the 2010s and early 2020s to estimate how the 2015 "Give Five" reform in advanced mathematics is related to subsequent participation in selective STEM programs. The analysis links two levels of administrative data: high-school Bagrut records and higher-education enrollment tables published by the CBS. The Bagrut data provide information on advanced-mathematics participation before and after the reform, while the higher-education data track the fields chosen by new university entrants from each locality.

5.1 Measuring exposure and constructing outcomes

The first step is to measure local exposure to the reform. Using school-level Bagrut data, I compute for each locality r and year t the share of students taking the five-unit mathematics exam. This share, denoted $\text{share5u}_{r,t}$, is then averaged separately over a pre-reform window and a post-reform window. Specifically, I define

$$\text{share5u}_r^{\text{pre}} = \frac{1}{2} \sum_{t \in \{2013,2014\}} \text{share5u}_{r,t}, \tag{1}$$

$$\text{share5u}_r^{\text{post}} = \frac{1}{4} \sum_{t \in \{2016,2017,2018,2019\}} \text{share5u}_{r,t}. \tag{2}$$

The difference between these two averages,

$$\text{Exposure}_r \equiv \Delta\text{share5u}_r = \text{share5u}_r^{\text{post}} - \text{share5u}_r^{\text{pre}}, \tag{3}$$

captures how much a locality expanded its participation in advanced math after the reform. Localities are classified into high, medium, and low exposure groups based on the realized distribution of $\Delta\text{share5u}_r$; in particular, I define a binary indicator HighExposure_r that equals one for localities in the upper part of the exposure distribution. In addition, the continuous exposure measure Exposure_r is retained, and later specifications use it directly to study dose–response patterns.

Each locality is merged with socio-economic characteristics from the CBS, such as population size and socio-economic cluster, which are used as controls in some cross-sectional specifications and as grouping variables in the heterogeneity analysis. These variables are measured prior to the outcome period.

Higher-education outcomes come from the CBS annual *Students in Higher Education* tables, which report the number of new first-degree students by field of study and by locality of residence at age 18. These data allow me to construct, for every locality r and year t, the outcome variable:

$$\text{STEM}_{r,t} = \frac{\text{new STEM students}_{r,t}}{\text{total new first-degree students}_{r,t}}. \tag{4}$$

The STEM category aggregates engineering and architecture, mathematics and computer science, physical sciences, and biological sciences, fields that are most directly affected by advanced-math preparation in high school. Secondary analyses, discussed in the results section, examine specific subfields such as computer science or engineering alone. Since university enrollment typically occurs several years after high school, the panel of higher-education outcomes is constructed for 2017 to 2024, which corresponds to the period in which cohorts that were exposed to the reform in high school plausibly appear in higher education.

5.2 Continuous exposure analysis: change-on-change design

Before moving to panel regressions, I begin with a simplified "change-on-change" exercise that examines whether localities more exposed to the Give-Five reform experienced larger raw increases in STEM entry. The idea is straightforward: if the reform's mechanism, expansion of five-unit mathematics, was indeed linked to greater STEM participation, we should observe a positive cross-sectional correlation between each locality's increase in advanced-math exposure and its subsequent rise in STEM enrollment. This step provides a transparent descriptive validation of the hypothesized channel before imposing any panel-based identifying assumptions.

For each locality r, I compute the pre- and post-reform averages of the STEM share:

$$\text{STEM}_r^{\text{pre}} = \frac{1}{3} \sum_{t \in \{2017,2018,2019\}} \text{STEM}_{r,t}, \tag{5}$$

$$\text{STEM}_r^{\text{post}} = \frac{1}{5} \sum_{t \in \{2020,2021,2022,2023,2024\}} \text{STEM}_{r,t}. \tag{6}$$

The baseline cross-sectional regression is

$$\Delta\text{STEM}_r = \alpha + \beta_1 \text{Exposure}_r + u_r, \tag{7}$$

estimated with population weights. This specification asks whether localities with larger realized exposure to the Give-Five reform experienced larger changes in STEM entry between the pre- and post-reform periods.

To probe the robustness of this relationship, I extend the model by including controls for pre-reform STEM levels, socio-economic status, and locality size. Let $\text{STEM}_r^{\text{pre}}$ denote the pre-period STEM share, SES_r a measure of socio-economic status based on the CBS cluster index, and Size_r the log of the 2015 population. The extended specification is

$$\Delta\text{STEM}_r = \alpha + \beta_1\text{Exposure}_r + \beta_2\text{STEM}_r^{\text{pre}} + \beta_3\text{SES}_r + \beta_4\text{Size}_r + u_r. \tag{8}$$

In addition, I estimate a version that includes fixed effects for the socio-economic cluster (for example, cluster-2015 categories) to soak up differences across clusters. The change-on-change regressions are descriptive rather than causal, and they are complemented by a weighted binned-scatter plot in which localities are grouped into bins of the exposure index, and the binned values of ΔSTEM_r are plotted against average exposure. Together, these tools provide an intuitive graphical and regression-based view of the association between advanced-math expansion and changes in STEM entry.

5.3 Initial difference-in-differences framework

The next step uses the panel dimension of the data to implement a two-way fixed-effects difference-in-differences (DiD) design. The goal is to test whether localities that were highly exposed to the expansion of five-unit mathematics experienced larger post-2019 increases in the share of entrants to STEM majors.

I estimate the following DiD specification:

$$\text{STEM}_{r,t} = \alpha_r + \gamma_t + \beta(\text{HighExposure}_r \times \text{Post}_t) + \varepsilon_{r,t}, \tag{9}$$

where $\text{STEM}_{r,t}$ is the STEM share among new first-degree students from locality r in year t, HighExposure_r is a 0/1 indicator derived from the exposure index Exposure_r, and Post_t equals 1 for 2020 to 2024 and 0 for 2017 to 2019. Locality fixed effects α_r net out time-invariant differences across places, such as geography, baseline income, and long-run educational culture. Year fixed effects γ_t capture common national shocks such as admissions changes, curriculum adjustments, macroeconomic conditions, or the COVID-19 pandemic. Standard errors are clustered at the locality level, and the regression is weighted by 2015 population.

In this setup, the coefficient β measures the average difference in the post-2019 change in STEM enrollment between high-exposure and low-exposure localities, controlling for persistent locality characteristics and national trends. This simple specification implements the textbook DiD under the parallel-trends assumption: absent the reform, high-exposure and comparison localities would have followed the same trend in STEM entry over the sample period.

5.4 Pre-trend test and trend-robust DiD

Because the parallel-trends assumption is central, I explicitly examine whether high-exposure and low-exposure localities followed similar trajectories before treated cohorts could affect higher-education outcomes. To do so, I estimate a pre-trend specification using only the years 2017 to 2019.

Let t index these pre-outcome years and let Trend_t be a normalized time variable that increases linearly over 2017, 2018, and 2019. The pre-trend regression is

$$\text{STEM}_{r,t} = \alpha_r + \gamma_t + \phi(\text{Trend}_t \times \text{HighExposure}_r) + \varepsilon_{r,t}, \qquad t \in \{2017, 2018, 2019\}. \quad (10)$$

This equation includes locality and year fixed effects, as before, and adds an interaction between the linear time trend and the high-exposure indicator. The coefficient ϕ indicates whether high-exposure localities exhibit a different pre-reform slope in STEM participation relative to other localities. This specification serves as a diagnostic test for the validity of the parallel-trends assumption in the simple DiD.

Motivated by the pre-trend evidence, I then augment the DiD specification to allow high-exposure and low-exposure localities to follow different linear trends over the entire outcome window. The trend-robust DiD model is

$$\text{STEM}_{r,t} = \alpha_r + \gamma_t + \beta(\text{HighExposure}_r \times \text{Post}_t) + \phi(\text{Trend}_t \times \text{HighExposure}_r) + \varepsilon_{r,t}, \quad (11)$$

estimated for 2017 to 2024 with the same weights and clustering. In this specification, the interaction $\text{Trend}_t \times \text{HighExposure}_r$ captures the group-specific linear trend, while β is interpreted as the additional post-2019 change in STEM share for high-exposure localities relative to what would be predicted by continuing their pre-period trend. This trend-adjusted DiD provides a more conservative benchmark for assessing whether there is evidence of a discrete break in STEM outcomes associated with the timing of the reform.

5.5 Socioeconomic inequality and continuous triple-difference analysis

A central question in evaluating the Give-Five reform is whether its impact on the STEM pipeline differed systematically across the socioeconomic distribution. If the reform functioned not only to expand the overall pool of advanced-math students but also to disproportionately benefit disadvantaged communities, one would expect stronger post-reform gains in STEM outcomes for low-SES localities. Identifying such differential effects in a panel setting with locality and year fixed effects is challenging, because many SES measures and exposure categories are time invariant and therefore absorbed by locality fixed effects. Standard triple-difference specifications that rely on categorical "high exposure" treatment groups suffer from limited within-cell variation and may yield imprecise estimates.

To address these concerns, I employ a continuous triple-difference (DDD) design that exploits the full cross-sectional variation in each locality's measured expansion of five-unit mathematics. Rather than comparing only high- versus low-exposure municipalities, I model exposure as a continuous index and interact it with both the post-period indicator and an indicator for low socioeconomic status.

Let $\text{LowSES}_r = 1$ for localities in CBS clusters 1 to 4 and $\text{LowSES}_r = 0$ otherwise. The estimating equation is

$$\text{STEM}_{r,t} = \alpha_r + \gamma_t + \beta_1(\text{Exposure}_r \times \text{Post}_t) + \beta_2(\text{Exposure}_r \times \text{Post}_t \times \text{LowSES}_r) + \varepsilon_{r,t}, \quad (12)$$

with observations weighted by 2015 population and standard errors clustered at the locality level. Here, β_1 captures the post-reform exposure–STEM slope among non-low-SES localities, while $\beta_1 + \beta_2$ captures the corresponding slope for low-SES localities. The coefficient β_2 therefore identifies whether disadvantaged municipalities experienced systematically larger or smaller gains in STEM participation for a given increase in advanced-math exposure, relative to more advantaged municipalities.

This continuous DDD approach embeds the equity question directly into the slope of the exposure–outcome relationship. By using the continuous exposure index rather than a binary "high exposure" dummy, it avoids throwing away variation and allows the analysis to make use of the full range of realized reform intensity when comparing SES groups. Locality fixed effects absorb all time-invariant characteristics, while year fixed effects absorb common shocks, so identification comes from differential changes across localities with different exposure levels and SES status over time.

5.6 Summary of the empirical strategy

Together, these models form a coherent empirical framework that mirrors the sequence implemented in the analysis. The continuous change-on-change regressions provide descriptive evidence on how changes in advanced mathematics are related to changes in STEM entry across localities. The difference-in-differences specifications use the panel structure to compare high-exposure and low-exposure localities over time, while explicitly probing and then relaxing the parallel-trends assumption through pre-trend tests and group-specific linear trends. The continuous triple-difference design adds a crucial equity dimension by asking whether the exposure–STEM relationship differs systematically between low-SES and non-low-SES localities once the reform has had time to operate through the educational pipeline. The next section presents the estimates from these specifications and interprets them in light of the Give-Five reform and the broader question of how advanced mathematics shapes the STEM pipeline.

6 Summary statistics

Panel A1 reports summary statistics for a balanced locality-year panel from 2017-2024. Each observation is a locality in a given year; outcomes are weighted by 2015 population so that large places contribute proportionally to averages. Our main outcome is the share of new undergraduates from each locality who enter STEM. On average, the population-weighted STEM share is 36.2 percentage. Timewise, the mean rises from 34.9 percentage in 2017-2019 to 37.0 percentage in 2020-2024, an increase of roughly 2.1 pp. We treat this as descriptive headroom rather than evidence of impact, but it shows the outcome is moving in a direction in which an effect of the math reform could plausibly appear. Localities are sizeable and heterogeneous (1,312 locality-years; mean 2015 population 214k;), which underscores the choice to weight by population. The "exposure index (pp 5u)" varies meaningfully across places, capturing how much the Give-5 initiative expanded 5-unit math locally; that cross-sectional variation is what lets us ask whether places with larger math expansions also see larger shifts into STEM.

Table 1: Panel A1 – Summary statistics (All)

	All		
Variable	**Mean**	**Std. Dev.**	**Obs.**
STEM share (overall)	36.2	7.6	1312
STEM share (2017–2019)	34.9	7.2	481
STEM share (2020–2024)	37.0	7.7	831
Exposure index (pp 5u)	0.053	0.04	1312
Population 2015	214 251	263 370	1312
SES cluster (2015)	4.971	2.368	1312
Log population 2015	11.514	1.319	1312

Notes: Weighted by 2015 population where applicable.

Panel A2, A3 below breaks the same statistics by socio-economic status (SES) and by exposure groups. By SES, three patterns stand out. First, levels differ sharply: high-SES localities average 40.7% STEM (SD 5.0 pp), mid-SES 39.3% (4.8 pp), and low-SES 31.1% (7.9 pp). The $\approx$ 9–10 pp high–low gap is present before the reform could influence college major choice (2017–2019: 38.8% vs. 30.1%) and persists afterward (2020–2024: 41.9% vs. 31.7%). Second, dispersion is much larger in low-SES areas, suggesting greater heterogeneity in underlying conditions and hinting that strict parallel trends may be fragile without additional controls. Third, the exposure index itself is stratified by SES (e.g., higher on average in high-SES places), which cautions against interpreting raw differences as causal.

The exposure split tells a consistent story. "High-exposure" localities average 40.4% STEM (SD 4.9 pp) compared with 34.5% (SD 7.8 pp) in "low-exposure" localities. The gap is visible already in 2017–2019 (38.3% vs. 33.5%) and both series rise post-2020 (41.6% vs. 35.1%). This alignment between exposure to advanced math and STEM participation is exactly the

relationship our analysis aims to test more rigorously. At the same time, because exposure correlates with pre-period levels and SES, simple pre/post contrasts could confound reform effects with persistent locality characteristics.

Table 2: Panel A2 – Summary statistics by SES group

	By SES group								
	High			Mid			Low		
Variable	**Mean**	**SD**	**Obs.**	**Mean**	**SD**	**Obs.**	**Mean**	**SD**	**Obs.**
STEM share (overall)	40.7	5.0	185	39.3	4.8	392	31.1	7.9	735
STEM share (2017–2019)	38.8	4.6	69	38.0	4.5	145	30.1	7.8	267
STEM share (2020–2024)	41.9	4.8	116	40.0	4.7	247	31.7	8.0	468
Exposure index (pp 5u)	0.10	0.029	185	0.059	0.026	392	0.026	0.033	735
Population 2015 (thousands)	190.4	173.9	185	150.1	89.2	392	290.1	372.8	735
SES cluster (2015)	8.06	0.237	185	6.154	0.877	392	2.396	0.839	735
Log population 2015	11.598	1.158	185	11.604	0.941	392	11.386	1.661	735

Notes: Weighted by 2015 population where applicable.

Table 3: Panel A3 – Summary statistics by exposure group

	By exposure group					
	High			Low		
Variable	**Mean**	**SD**	**Obs.**	**Mean**	**SD**	**Obs.**
STEM share (overall)	40.4	4.9	305	34.5	7.8	1007
STEM share (2017–2019)	38.3	4.8	113	33.5	7.6	368
STEM share (2020–2024)	41.6	4.6	192	35.1	7.9	639
Exposure index (pp 5u)	0.098	0.024	305	0.034	0.028	1007
Population 2015 (thousands)	190.5	143.7	305	224.2	299	1007
SES cluster (2015)	7.242	1.324	305	4.021	2.039	1007
Log population 2015	11.709	1.122	305	11.433	1.386	1007

Notes: Weighted by 2015 population where applicable.

Taken together, Panels A1-A2 motivate the empirical choices used later. We estimate models with two-way fixed effects (locality and year) and population weights to net out time-invariant locality traits and common shocks. We explicitly probe pre-period differences and, where indicated, include group-specific trends. In parallel, we move beyond binary treatment and estimate continuous-exposure DiD specifications that relate changes in STEM to measured growth in 5-unit math. The objective is modest and testable: assess whether the post-2019 widening in STEM associated with Give-5 survives once fixed differences, SES composition, and plausible deviations from parallel trends are taken into account.

6.1 Time-Series Patterns in STEM Participation

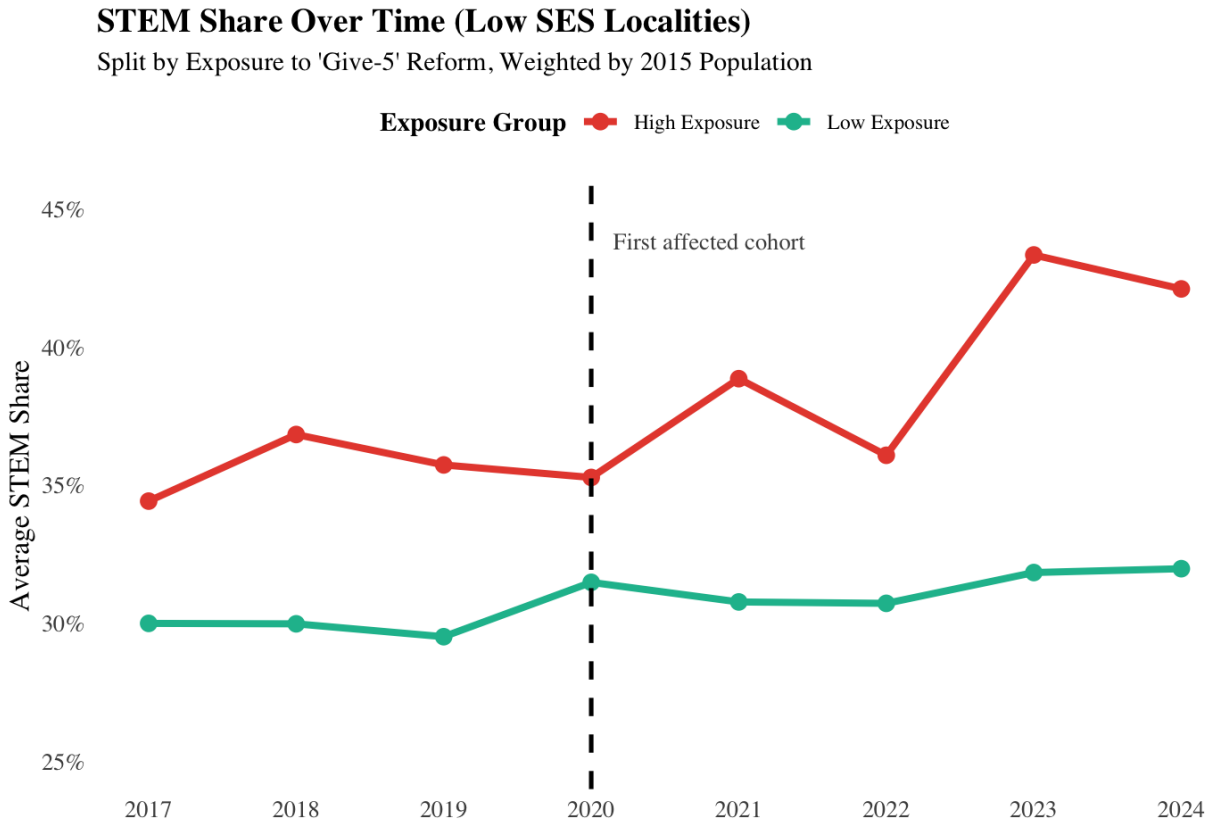


Figure 2: STEM Share Over Time (Low SES Localities)

Figure 2A presents the evolution of STEM participation across Israeli localities between 2017 and 2024, weighted by population to reflect aggregate trends. Both panels provide suggestive evidence of patterns that motivate our later empirical analysis.

Panel 2A focuses on low-SES localities, split by their exposure to the Give-5 reform, while Panel 2B traces average STEM shares by socio-economic status. the gap between high and low-exposure localities appears to widen after 2020, the first year when cohorts potentially affected by the high-school math reform reached university. Among low-SES localities, those with higher exposure to advanced mathematics show a visibly stronger upward trajectory, rising from roughly 36% to above 43% STEM share, whereas low-exposure localities remain relatively flat around 32%. While this pattern is consistent with an effect of the reform, it may also reflect pre-existing differences in local emphasis on math and science; the figure should therefore be viewed as descriptive rather than causal.

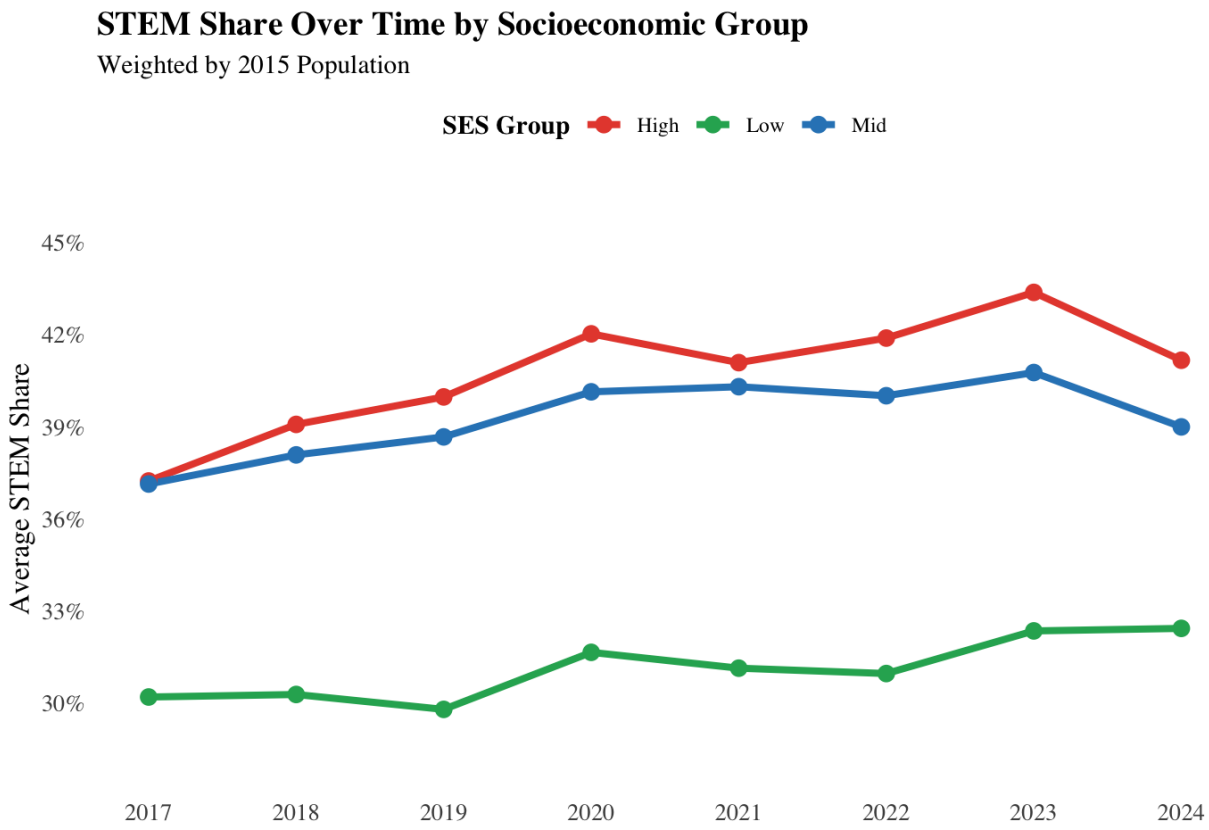


Figure 3: STEM Share Over Time (Low SES Localities)

Panel 1B shows how these trends interact with socio-economic status. STEM participation is consistently higher in high-SES localities (around 40–44%) than in low-SES ones (about 30–32%), with mid-SES localities in between. All groups show modest growth over time, but the relative gaps remain stable. This persistence suggests that the Give-5 reform did not immediately narrow socio-economic disparities in STEM entry, though it may have encouraged general growth in participation across the board. The larger year-to-year fluctuations observed in the low-SES series point to greater heterogeneity and raise the possibility of non-parallel pre-trends, an issue our subsequent econometric analysis will explicitly test.

Taken together, these figures provide preliminary, visual indications that exposure to the high-school math expansion correlates with higher and rising STEM participation, while socio-economic differences remain substantial. The next section formalizes these observations using a two-way fixed-effects difference-in-differences design, assessing whether the apparent post-2019 divergence in STEM enrollment by exposure level reflects a genuine causal impact of the Give-5 reform or simply the continuation of underlying trends.

6.2 Continuous Exposure Analysis

Before moving to the full difference-in-differences specification, we begin with a simplified "change-on-change" exercise that examines whether localities more exposed to the Give-5 reform experienced larger raw increases in STEM entry. The idea is straightforward: if the reform's mechanism, expansion of five-unit mathematics, was indeed linked to greater STEM participation, we should observe a positive cross-sectional correlation between each locality's increase in advanced-math exposure and its subsequent rise in STEM enrollment. This step provides a transparent descriptive validation of the hypothesized channel before imposing any panel-based identifying assumptions.

Table 4 summarizes these regressions, where the dependent variable is the population-weighted change in the share of new STEM entrants between the pre- (2017–2019) and post-reform (2020–2024) periods. The main regressor is the locality-specific Exposure Index, measuring how much "Give-5" expanded advanced-math participation. Column A reports a simple bivariate regression, Column B adds controls for baseline STEM levels, socio-economic status, and locality size, and Column C further includes fixed effects for the socio-economic cluster (cluster 2015). Across all specifications, the coefficient on exposure remains positive, highly significant, and remarkably stable in magnitude. In the preferred specification (Column C), a one-unit increase in the exposure index predicts roughly a 0.30 rise in the STEM share, corresponding to an increase of about 3 percentage points for a 0.10-point change in exposure or roughly 1.6 pp for a one-standard-deviation difference. The model's explanatory power rises from $R^2 = 0.08$ in the simple model to 0.27 once controls and fixed effects are added, suggesting that the relationship is robust to observable locality characteristics.

Table 4: Continuous Exposure: ΔSTEM (post–pre) on Exposure Index

	Simple (A)	+Controls (B)	+Cluster FE (C)
Exposure Index	0.197***	0.322***	0.295**
	(0.036)	(0.063)	(0.080)
Num. Obs.	164	164	164
R^2	0.075	0.177	0.266
Adjusted R^2	0.069	0.151	0.213
Within R^2			0.208
Adj. Within R^2			0.192
Effect for +0.10 exposure	2.0%	3.2%	3.0%
Effect for +1 SD exposure	1.1%	1.7%	1.6%

Notes: Dependent variable is the change in the share of new STEM entrants between 2017–2019 and 2020–2024, weighted by 2015 population. Robust standard errors in parentheses. + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Figure 4 below provides a visual counterpart through a weighted binned-scatter plot of the change in STEM share against the exposure index. Each point represents an equally weighted group of localities, and the fitted line shows the clear positive slope underlying the regression estimates. The upward trend confirms that, on average, communities with stronger implementation of the Give-5 reform saw larger raw increases in STEM participation. These patterns are descriptive, not causal, but they are informative. The results indicate that exposure to advanced mathematics is strongly correlated with subsequent gains in STEM entry, even after adjusting for pre-existing levels and socio-economic composition. At the same time, because exposure is not randomly assigned and may co-vary with underlying trends, such as differential improvements in schooling or student composition, the correlation cannot be interpreted as a treatment effect. In the next stage, we therefore estimate a two-way fixed-effects difference-in-differences model that explicitly controls for time-invariant locality characteristics and common year shocks, and later allow for group-specific trends to address the potential non-parallel dynamics we have already suspected in the descriptive analysis.

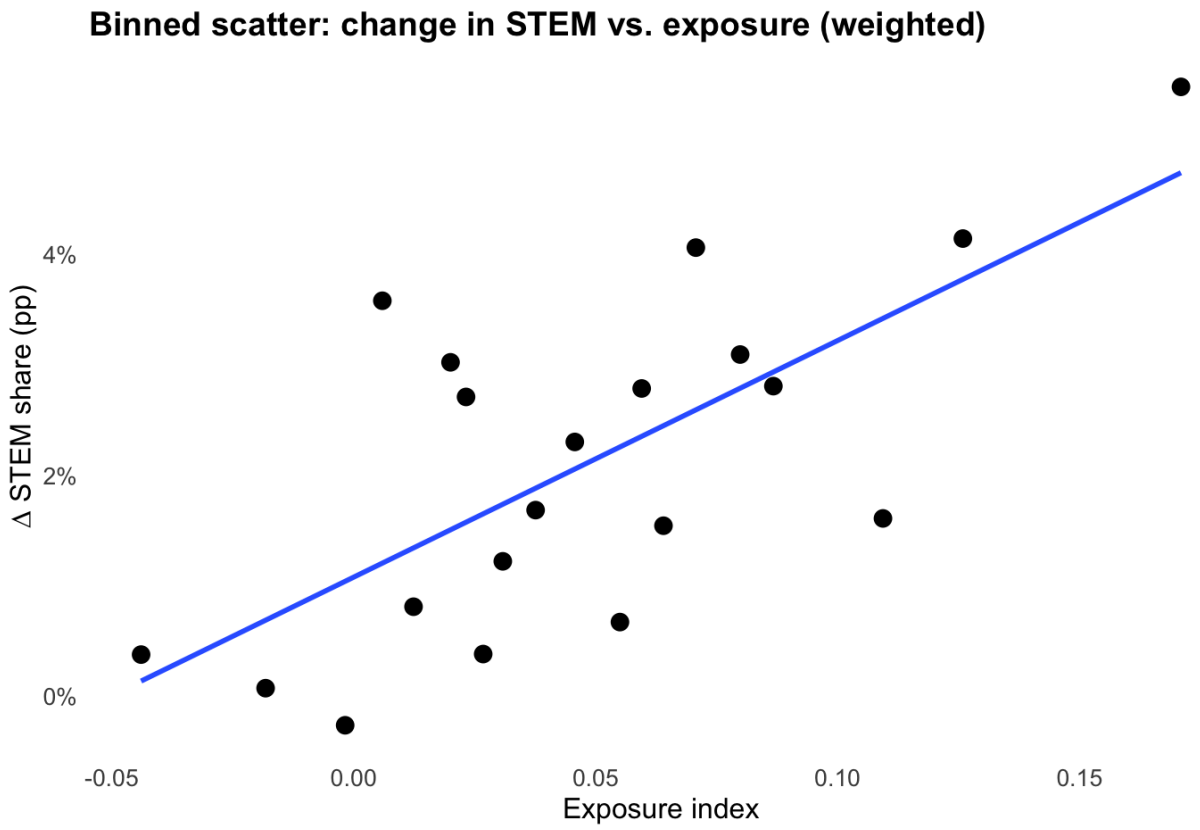


Figure 4: Change in STEM vs. exposure (weighted)

7 Results and Discussion

7.1 Initial Regression – design, rationale, and interpretation

Table 5: Initial Regression Estimates

	Simple DiD	Pre-Trend Test (2017–2019)	DiD + Trend Control
High × Post	0.016***		0.001
Trend × High (pre)		0.011**	
Num. Obs.	1309	478	1309
R^2	0.842	0.873	0.843
Adjusted R^2	0.818	0.807	0.818

Notes: Outcome: STEM share (weighted by 2015 population). + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

My goal is to test whether localities that were highly exposed to the "Give-5" expansion of 5-unit mathematics experienced larger post-2019 increases in the share of entrants to STEM majors. I begin with a transparent two-way fixed-effects difference-in-differences design, weighting observations by 2015 population so that larger localities have proportional influence. The dependent variable is the STEM share among new undergraduates from locality r in year t. The key regressor is the interaction $\text{HighExposure}_r \times \text{Post}_t$, where HighExposure_r is a 0/1 indicator derived from our exposure measure and Post_t equals 1 for 2020–2024, the first years in which cohorts plausibly affected by Give-5 appear in higher education. All specifications include locality fixed effects (netting out time-invariant differences across places) and year fixed effects (capturing common national shocks, e.g., admissions changes or COVID), with standard errors clustered by locality.

Column 1, Simple DiD: The coefficient on High × Post is 0.016 with a standard error of 0.004 ($p < 0.001$). Because the outcome is in proportion units, this maps to a +1.6 percentage-point increase in STEM share, on average, for high-exposure localities relative to low-exposure localities after 2019. The estimate is precise (SE 0.4 pp) and the sample is large ($N = 1{,}309$ locality-years). This specification implements the textbook DiD under the parallel-trends assumption: absent the reform, treated and control localities would have followed the same trend.

Column 2, Pre-trend test (2017–2019): To probe the parallel-trends assumption directly, we restrict the sample to 2017–2019 and regress STEM share on a linear time trend t, a treatment indicator, and their interaction $t \times$ High (with the same locality and year fixed effects). The coefficient on Trend × High (pre) is 0.011 (SE 0.003, $p < 0.01$). Interpreted literally, high-exposure localities were already trending 1.1 pp per (year-unit) faster than low-exposure localities before treated cohorts could affect college majors. This constitutes evidence against strict parallel pre-trends, at least in a linear sense, and cautions against

taking simple DiD at face value.

Column 3, DiD with trend control: I therefore augment the DiD with a group-specific linear trend, adding $t \times$ High to the full 2017–2024 panel. Once I allow the high-exposure group to continue its pre-period slope, the post-2019 interaction shrinks to 0.001 (SE 0.006), i.e., economically small and statistically indistinguishable from zero. This pattern is consistent with an explanation in which the simple DiD effect was picking up a continuation of existing differential dynamics rather than a new post-Give-5 shift.

The initial estimates indicate that model fit is high ($R^2 \approx 0.84$ in Columns 1 and 3), which is expected given the absorbing power of locality and year fixed effects; these soak up persistent cross-place differences (e.g., school quality, STEM culture, proximity to universities) and nationwide shocks (curriculum changes, labor-market conditions, the pandemic), so identification comes from within-locality changes over time. Under the simple DiD that assumes parallel trends, the interaction "High $\times$ Post" is positive and precise, about +1.6 percentage points with a standard error near 0.004, suggesting that high-exposure places rise more after 2019. Yet once we allow for a differential pre-period slope (Column 3), the standard error increases only modestly (to ~ 0.006) while the point estimate collapses to ~ 0, implying that the Column-1 effect was largely picking up pre-existing momentum rather than a post-reform break. These are preliminary results and the pre-trend adjustment is simple and linear; non-linear dynamics, measurement of exposure at the locality level, and other evolving features (e.g., migration or school openings) could matter.

Bottom line: the simple DiD suggests a +1.6 pp association, but under a trend-robust specification the aggregate post-2019 effect is indistinguishable from zero, motivating richer exposure measures and further robustness before drawing firm conclusions about Give-5's causal impact on STEM entry.

7.2 Socioeconomic Inequality and the STEM Pipeline: A Continuous Triple-Difference Analysis

A central question in evaluating the Give-Five reform is whether its impact on the STEM pipeline differed systematically across the socioeconomic distribution. If the reform functioned not only to expand the overall pool of advanced-math students but also to disproportionately benefit disadvantaged communities, one would expect stronger post-reform gains in STEM outcomes for low-SES localities. Identifying such differential effects is challenging in panel settings with saturated fixed effects, because many SES measures and exposure categories are time-invariant and therefore absorbed by locality fixed effects. Standard triple-difference specifications that rely on categorical "high exposure" treatment groups suffer from limited within-cell variation and yield imprecise estimates.

To overcome these limitations, I employ a continuous triple-difference (DDD) design that exploits the full cross-sectional variation in each locality's measured expansion of five-unit mathematics. Rather than comparing only high- versus low-exposure municipalities, I model

exposure as a continuous index, defined as the shift in 5-unit participation from the pre-reform (2013–2014) to the post-reform (2016–2019) period, and interact it with a post-period indicator and a low-SES indicator. The identifying assumption is that, conditional on locality and year fixed effects, differential changes in STEM outcomes across localities with varying exposure intensities capture the causal effect of Give-Five operating through advanced-math expansion.

Formally, I estimate:

$$\text{STEM}_{rt} = \beta_1(\text{Exposure}_r \times \text{Post}_t) + \beta_2(\text{Exposure}_r \times \text{Post}_t \times \text{LowSES}_r) + \alpha_r + \gamma_t + \varepsilon_{rt},$$

where r indexes localities and t years; α_r and γ_t are locality and year fixed effects; and observations are weighted by 2015 population and clustered at the locality level. Here, β_1 captures the post-reform exposure–STEM slope among non-low-SES localities, while $\beta_1 + \beta_2$ captures the corresponding slope for low-SES localities. The coefficient β_2 therefore identifies whether disadvantaged municipalities experienced systematically larger (or smaller) gains in STEM participation for a given increase in advanced-math exposure.

Table 6: Continuous DDD Regression: Exposure × Post and Triple Interaction

	Estimate	**Std. Error**
Exposure × Post	0.202***	0.054
Exposure × Post × LowSES	0.004	0.068
Num. Obs.	1,309	
RMSE	0.030	
Adj. R^2	0.818	
Within R^2	0.016	
Fixed effects	Locality and Year	
SEs clustered	Locality level	
Weights	2015 population	

Notes: OLS estimates with locality and year fixed effects. Standard errors clustered at the locality level. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 6 reports the results of this continuous DDD specification. The exposure–post interaction is large and highly statistically significant:

$$\hat{\beta}_1 = 0.202 \ (\text{SE} = 0.054, \ p < 0.001).$$

This coefficient implies that a 0.10-point increase in exposure, e.g., a ten-percentage-point rise in the share of students taking five units- predicts roughly a 2.0-percentage-point increase in the share of new university entrants who pursue STEM fields. The result is both economically meaningful and consistent with the reform's stated mechanism: localities that expanded advanced mathematics more aggressively subsequently supplied more STEM students to higher education.

The triple-interaction term, however, is nearly zero and estimated with great precision:

$$\hat{\beta}_2 = 0.0036 \text{ (SE} = 0.0676,\ p = 0.957).$$

The point estimate is an order of magnitude smaller than the main effect and statistically indistinguishable from zero. This indicates that low-SES localities do not exhibit a differential exposure–STEM gradient: for a given increase in five-unit participation, disadvantaged municipalities do not experience larger (or smaller) increases in STEM entry than their more advantaged counterparts. In substantive terms, the Give-Five reform strengthened the math-to-STEM pipeline across the board, but it did not alter the slope of this pipeline for low-SES communities.

Taken together, the continuous DDD results reveal a clear and coherent pattern. The reform's intensity measured by the realized increase in advanced-math participation, strongly predicts subsequent STEM enrollment, confirming the plausibility of the proposed mechanism. However, the mechanism appears distributionally neutral: it neither narrowed nor widened SES gaps in the transition to STEM. Localities starting from weaker socioeconomic positions did not disproportionately benefit from additional exposure to the reform; the gains accrued uniformly across the SES distribution. This finding aligns with the descriptive evidence presented earlier, where high-exposure localities consistently outperform low-exposure localities within each SES tier, but the relative SES gaps remain largely stable over time.

In sum, while Give-Five effectively expanded the supply of STEM-bound students, the continuous DDD estimates suggest that the reform did not function as an equalizing force across socioeconomic strata. Rather, its benefits appear to have been broadly proportional to exposure, without systematically favoring the most disadvantaged communities.

8 Conclusions

This paper followed the Give-Five reform from high school into higher education and asked two linked questions: did local expansions in five-unit mathematics raise entry into selective STEM programs, and did disadvantaged communities benefit more? Using linked Bagrut and CBS enrollment data for 2013–2024, I constructed a locality–year panel and an exposure index measuring the increase in the five-unit share between 2013–2014 and 2016–2019. The outcome is the STEM share among new first-degree students from each locality between 2017 and 2024.

The continuous exposure analysis shows a clear and stable association. In the change-on-change regressions, the dependent variable is the population-weighted change in the STEM share between 2017–2019 and 2020–2024, and the key regressor is the exposure index. In the simple bivariate specification, a one-unit increase in exposure is associated with a 0.197 rise in the STEM share, which corresponds to about 2.0 percentage points for a 0.10 increase in exposure. With controls for baseline STEM, socio-economic status, and locality size, the coefficient rises to 0.322, and in the preferred specification with socio-economic cluster fixed

effects it is 0.295. These latter estimates imply that a 0.10 increase in exposure is associated with roughly a 3.0 percentage-point increase in STEM share, and that a one-standard-deviation difference in exposure (about 0.055) corresponds to roughly 1.6 percentage points. The R^2 of the cross-sectional models rises from 0.08 to 0.27 as controls are added, indicating that exposure retains substantial explanatory power even after accounting for observable differences across localities.

Panel regressions refine this picture. A simple two-way fixed-effects DiD, comparing high-exposure and lower-exposure localities before and after 2020, yields a coefficient of about 0.016 on *High* × *Post*. Interpreted in levels, this corresponds to a 1.6 percentage-point higher STEM share in high-exposure municipalities in the post-2019 period, on top of common time effects and fixed locality characteristics. However, a pre-trend test that interacts a linear time trend with the high-exposure indicator over 2017–2019 produces a slope difference of about 0.011 per year, showing that high-exposure places were already gaining STEM share faster before treated cohorts reached university. When the main DiD specification is augmented with this group-specific trend, the *High* × *Post* coefficient shrinks to roughly 0.001 with a standard error around 0.006, which is statistically and economically negligible. This pattern suggests that the simple DiD contrast was largely capturing a continuation of pre-existing momentum rather than a sharp post-reform break.

The continuous triple-difference model provides the cleanest numerical summary of how exposure and socio-economic status interact. The coefficient on $\text{Exposure}_r \times \text{Post}_t$ is about 0.202 with a standard error of roughly 0.054. This implies that a ten-percentage-point increase in five-unit participation is associated with about a 2.0 percentage-point increase in the STEM share of new university entrants, conditional on locality and year fixed effects. In contrast, the triple interaction $\text{Exposure}_r \times \text{Post}_t \times \text{LowSES}_r$ is about 0.004 with a standard error near 0.068, statistically indistinguishable from zero. Numerically, this means that the slope of STEM on exposure in low-SES localities differs from that in other localities by at most a few tenths of a percentage point per 0.10 of exposure, well within the margin of sampling error.

Taken together, the numbers support a clear conclusion. At the locality level, stronger implementation of Give-Five, as captured by realized increases in five-unit mathematics participation, is associated with sizeable increases in the share of graduates who enter STEM fields. A 0.10 rise in exposure predicts gains in STEM share on the order of 2 to 3 percentage points, which is meaningful given baseline STEM shares in the low 30s. At the same time, the continuous DDD estimates show no detectable difference in this exposure–STEM gradient between low-SES and higher-SES municipalities. The reform appears to have shifted the overall STEM pipeline upward rather than tilting it toward disadvantaged communities. From a policy perspective, this suggests that expanding advanced mathematics can effectively increase the aggregate supply of STEM-bound students, but that additional, more targeted tools are likely required if the goal is to close socio-economic gaps in who ultimately reaches those high-return fields.

9 Bibliography

References

[1] Knesset Research and Information Center. 2016. "Main Reforms in the Ministry of Education During the Term of the Twentieth Knesset." *Knesset Research and Information Center Report*. https://shorturl.at/KtDn6.

[2] Maagan, David, and Noam Zussman. 2019. "Assessing the Effectiveness of the 'Scientific-Technological Reserves' and 'Give Five' Programs." *Bank of Israel Discussion Paper 2019.10*. https://www.boi.org.il/media/g04p3loy/dp201910e.pdf.

[3] Council for Higher Education (CHE). 2024. "Statistical Data Portal: Students and Freshmen by Field and Institution Type." *Council for Higher Education Data Portal*. https://che.org.il.

[4] Blass, Nachum. 2020. "Achievements and Gaps: The Status of the Israeli Education System." *Taub Center for Social Policy Studies in Israel*. https://www.taubcenter.org.il.

[5] Israel Democracy Institute (IDI). 2022. "STEM Pipeline and 5-Unit Math Analysis, 2019–2022." *IDI Research Report*. https://www.idi.org.il.

[6] State Comptroller of Israel. 2024. "Activities of Local Authorities for the Integration of Israelis of Ethiopian Origin into Society." *State Comptroller Report*. https://www.mevaker.gov.il.

[7] Ynet. 2023. "FOI 'Bagrut Map': School/Locality-Level Bagrut Results Since 2013." *Ynet FOI Database*. https://bagruyot.ynet.co.il.

[8] Government of Israel. 2020. "Give Five: The National Program for Strengthening Mathematics Studies." *Ministry of Education Publication*. https://www.gov.il/he/pages/give-me-five-math.